

6 ξ - α NET Spectral Components Search Space

As demonstrated in Supplementary Materials A, the ξ - α NET generative model predicts the sensor-level cross-spectrum of EEG signals at frequency ω according to the forward model

$$\Sigma_{\omega}^v = \sigma_{\omega}^2 \mathbf{I}_{N_c} + \mathbf{T}_{\omega} \text{diag}(\boldsymbol{\xi}_{\omega} + \boldsymbol{\alpha}_{\omega}) \mathbf{T}_{\omega}^{\dagger},$$

where $\sigma_{\omega}^2 \mathbf{I}_{N_c}$ represents sensor noise, $\mathbf{T}_{\omega}$ denotes the frequency-dependent forward operator, and the diagonal term models voxelwise latent spectral activity, composed of a smooth aperiodic process $\boldsymbol{\xi}_{\omega}$ and a peaked oscillatory process $\boldsymbol{\alpha}_{\omega}$. To estimate the latent spectral parameters from empirical observations, we first project the measured cross-spectrum $\mathbf{S}_{\omega}$ into the latent space by applying a minimum-norm back-projection. Specifically, we compute

$$\mathbf{s}_{\omega} = \text{diag}(\mathbf{T}_{\omega}^+ \mathbf{S}_{\omega} (\mathbf{T}_{\omega}^+)^{\dagger}),$$

where $\mathbf{T}_{\omega}^+$ denotes the Moore–Penrose pseudoinverse of the forward operator. The resulting $\mathbf{s}_{\omega} \in \mathbb{R}^{N_v \times 1}$ contains the estimated power spectra for each voxel at frequency ω . It is important to note that this latent space projection is distinct from classical source localization: it represents the domain where the underlying Xi and Alpha processes are assumed to be generated.

Following the latent projection, we proceed to voxel-wise estimation of the Xi–Alpha parameters. For each voxel independently, the aperiodic parameters $\mathbf{e} = (e_1, e_2, e_3)$ and the alpha parameters $\mathbf{a} = (a_1, a_2, a_3, a_4)$ are obtained by solving the nonlinear least-squares problem

$$(\hat{\mathbf{e}}, \hat{\mathbf{a}}) = \arg \min_{\mathbf{e}, \mathbf{a}} \sum_{\omega} \|\log \mathbf{s}_{\omega} - \log(\boldsymbol{\xi}_{\omega} + \boldsymbol{\alpha}_{\omega})\|_2^2.$$

The logarithmic transformation in the cost function balances the contributions of low- and high-power frequencies and stabilizes the optimization numerically, although it preserves the nonlinearity of the mapping from parameters to spectra. The search space for the voxelwise optimization is carefully constructed to ensure physiological plausibility and robust convergence. The baseline term e_1 is fixed to the first value of the latent spectrum, $e_1 = \mathbf{s}_{\omega}(1)$, reflecting the power at the lowest frequency. The remaining parameters are randomly initialized within empirically motivated bounds: e_2 is initialized uniformly between 0 and 1, e_3 between 1 and 2.5, a_1 between 0.5 and 1 times the maximum value of $\mathbf{s}_{\omega}$, a_2 between 0 and 0.5, a_3 between 2 and 5, and a_4 between 7 and 13 Hz. The optimization is constrained within the bounds $e_1, e_2, e_3, a_1, a_2, a_3 > 0$ and $7 \leq a_4 \leq 13$ Hz, ensuring that the alpha peak remains within the expected physiological range [15, 14]. To mitigate the risk of local minima due to the nonconvexity of the problem, the voxelwise fitting procedure is repeated with 30 different random initializations for each voxel. For each initialization, the optimization is performed independently, and the parameter set that achieves the lowest residual error is selected as the voxelwise estimate $(\hat{\mathbf{e}}, \hat{\mathbf{a}})$. After fitting across all voxels, the variability of the parameter estimates is quantified by computing the empirical variance of each parameter across the voxel ensemble. This variance reflects the natural dispersion of spectral characteristics over the cortical surface. The final global search space for subsequent group-level or hierarchical modeling steps in the ξ - α NET framework is then defined by scaling this voxelwise variance by a factor of two:

$$\text{Search Space Width} = 2 \times \text{Var}(\hat{\mathbf{e}}, \hat{\mathbf{a}}).$$

Thus, the search space becomes data-adaptive, ensuring that the global optimization remains both flexible and constrained according to the physiological variability captured during the voxelwise estimation.

7 Selection of the Regularization Space

The estimation of the parameters in the ξ - α NET model is formulated as a maximum a posteriori (MAP) optimization problem, as discussed in Supplementary Material A. This problem is solved using a Stochastic FISTA algorithm, which relies on a proximal gradient method. A key step in this optimization procedure is the selection of an appropriate regularization space, determined by the parameters λ_1 , λ_2 , and λ_3 . To define this space rigorously, we seek the minimum regularization values such that, for any λ exceeding these thresholds, the proximal gradient method produces a trivial solution where all parameters are zero [1, 6]. Formally, the condition for this regime is expressed as

$$\text{prox}_g(\nabla_{\mathbf{x}} f(\mathbf{0}, \mathbf{w})) = \mathbf{0},$$

where $f(\mathbf{x}, \mathbf{w})$ denotes the smooth part of the MAP objective, $g(\mathbf{x})$ denotes the nonsmooth regularization term, $\mathbf{x} = (\text{vec}_r(\mathbf{e})^\top, \text{vec}_r(\mathbf{a})^\top, (\boldsymbol{\sigma}^2)^\top)^\top \in \mathbb{R}^{7N_v + N_\omega}$ collects the parameters to be estimated, and $\mathbf{w}$ represents fixed structural weights. Decomposing the regularization into its separate components, this global condition expands into three independent proximal operator conditions, one for each group of parameters: the noise variances $\boldsymbol{\sigma}^2$, the Xi parameters $\mathbf{e}$, and the Alpha parameters $\mathbf{a}$. For the variance parameters $\boldsymbol{\sigma}^2$, the regularization is a squared ℓ_2 norm, $\lambda_1 \|\cdot\|_2^2$. The proximal operator for this penalty has a closed-form solution given by

$$\text{prox}_{\lambda_1 \|\cdot\|_2^2}(v) = \frac{1}{1 + 2\lambda_1} v,$$

where $v = \nabla_{\boldsymbol{\sigma}^2} f(\mathbf{0}, \mathbf{w})$ denotes the gradient evaluated at the origin. Requiring that the proximal operator yields zero implies that $\frac{1}{1+2\lambda_1} v = 0$. Since the gradient is in general nonzero, the only way to satisfy this condition is to ensure that the scaling factor annihilates v , leading to the threshold

$$\lambda_1^* = \|\nabla_{\boldsymbol{\sigma}^2} f(\mathbf{0}, \mathbf{w})\|_2.$$

Turning to the Xi parameters $\mathbf{e}$, the regularization is given by the $\ell_{2,1}$ norm applied across the rows of the parameter matrix $\mathbf{e} \in \mathbb{R}^{N_v \times 3}$, where N_v is the number of voxels. The $\ell_{2,1}$ norm is defined as

$$\|\mathbf{e}\|_{2,1} = \sum_{j=1}^{N_v} \|\mathbf{e}_j\|_2,$$

where $\mathbf{e}_j$ denotes the j -th row of $\mathbf{e}$. The proximal operator for this penalty acts row-by-row via a block soft-thresholding rule:

$$\left(\text{prox}_{\lambda_2 \|\cdot\|_{2,1}}(V)\right)_j = \left(1 - \frac{\lambda_2}{\|V_j\|_2}\right)_+ V_j,$$

where $V = \nabla_{\mathbf{e}} f(\mathbf{0}, \mathbf{w})$ is the gradient matrix and $(\cdot)_+$ denotes the positive part operator. Setting the proximal operator to zero requires that for each row j , either the gradient norm $\|V_j\|_2$ is smaller than or equal to λ_2 , or the soft-thresholding reduces V_j to zero. Therefore, to ensure that all rows are simultaneously zeroed out, λ_2 must be greater than or equal to the maximum ℓ_2 norm across all rows of the gradient. This yields the threshold

$$\lambda_2^* = \max_j \left\| (\nabla_{\mathbf{e}} f(\mathbf{0}, \mathbf{w}))_j \right\|_2.$$

The same argument applies directly to the Alpha parameters $\mathbf{a} \in \mathbb{R}^{N_v \times 4}$, which are regularized using an $\ell_{2,1}$ penalty across their rows. Thus, the corresponding threshold is given by

$$\lambda_3^* = \max_j \left\| (\nabla_{\mathbf{a}} f(\mathbf{0}, \mathbf{w}))_j \right\|_2.$$

Collecting these results, the critical regularization thresholds ensuring trivial (zero) solutions are given by

$$\lambda_1^* = \|\nabla_{\boldsymbol{\sigma}^2} f(\mathbf{0}, \mathbf{w})\|_2, \quad \lambda_2^* = \max_j \left\| (\nabla_{\mathbf{e}} f(\mathbf{0}, \mathbf{w}))_j \right\|_2, \quad \lambda_3^* = \max_j \left\| (\nabla_{\mathbf{a}} f(\mathbf{0}, \mathbf{w}))_j \right\|_2.$$

These thresholds define the maximum regularization strengths allowable for each parameter group. Subsequently, regularization paths for model selection are constructed by defining grids of λ values below these critical values, ensuring that non-trivial, interpretable solutions are obtained during optimization. This construction of the regularization space guarantees that the model achieves a balance between accurate data fitting and sparsity-enforced interpretability.

8 Automatic selection of data-driven thresholds by Expectation Maximisation

The threshold-selection procedure begins with the one-dimensional sample $V = \{v_i\}_{i=1}^N$, which, in the context of the main text, represents the voxel-wise power of the SC term. All undefined entries are discarded, the vector

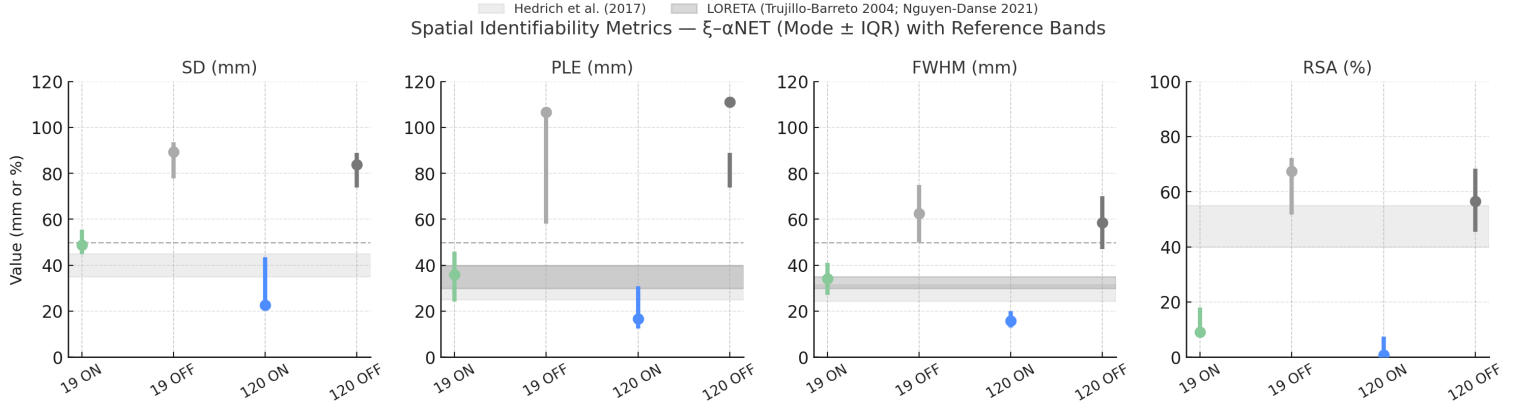


Figure 1: **Comparison of spatial identifiability metrics between ξ - α NET and hdEEG reference bands.** This figure presents the same spatial identifiability results shown in Fig. 3 (a-b) in the paper, but here the ξ - α NET estimates are directly compared against the reference intervals reported by Hedrich *et al.* [9] and other hdEEG studies (LORETA, dSPM, MNE, sLORETA; Nguyen-Danse *et al.* [13]). Even with only 19 electrodes, ξ - α NET achieves spatial dispersion (SD) and point-spread metrics (PLE, FWHM) within the benchmark range for 256-channel hdEEG systems, while yielding markedly lower RSA values—indicating a substantial reduction of spurious activity. With 120 channels, all metrics (SD, PLE, FWHM, RSA) improve further, fully matching or surpassing the hdEEG reference bands.

is flattened, and values whose magnitude falls below 10^{-10} are removed to prevent numerical pathologies. The cleaned data are then modelled as a two-component Gaussian mixture

$$p(v | \Theta) = \sum_{k=1}^2 \pi_k \mathcal{N}(v | \mu_k, \sigma_k^2), \quad \Theta = \{\pi_k, \mu_k, \sigma_k^2\}_{k=1}^2,$$

where $\pi_k > 0$ and $\pi_1 + \pi_2 = 1$ are the mixing proportions, μ_k the component means and σ_k^2 the variances. Maximum-likelihood estimates $\hat{\Theta}$ are obtained with the expectation–maximisation (EM) algorithm, which is restarted twenty times from random initial conditions; the solution with the highest final log-likelihood is retained. To avoid singular covariances, a small ridge constant 10^{-3} is added to each estimated variance, and iteration stops when the relative improvement of the log-likelihood drops below 10^{-3} or after one thousand iterations, whichever occurs first. If EM fails to converge—an eventuality that can arise when the empirical distribution is nearly unimodal—the method falls back to the empirical 90th percentile of V . Once the mixture has been fitted, the two Gaussians are labelled so that $\mu_1 < \mu_2$ and their variances are denoted σ_1^2 and σ_2^2 . The threshold is defined as the unique abscissa t^* lying between μ_1 and μ_2 at which the two unweighted densities are equal. Equating these densities yields the transcendental condition

$$\frac{(t - \mu_1)^2}{2\sigma_1^2} - \frac{(t - \mu_2)^2}{2\sigma_2^2} = \ln(\sigma_2/\sigma_1), \quad (\text{E.1})$$

which reduces to the midpoint $(\mu_1 + \mu_2)/2$ when the variances coincide. In the general, unequal-variance case, the equality of densities defines a smooth scalar function $f(t)$ whose root is sought numerically. Starting from the interval $[\mu_1, \mu_2]$, the endpoints are expanded symmetrically by successive multiples of the corresponding standard deviations until the function changes sign, after which a bracketing root-finder isolates the unique solution t^* . If a sign change cannot be produced within twenty expansions, the algorithm again reverts to the 90-th percentile fallback. The resulting value

$$t^* = \arg\{f(t) = 0\}, \quad (\text{E.2})$$

marks the point where the likelihood of membership in either component is equal under equal variances, or where the instantaneous contribution of the two components to the overall density is identical under unequal variances. This intersection, therefore, provides a statistically principled, data-adaptive demarcation between the low-power and high-power regimes of SC. Because the procedure requires no external tuning and accommodates both symmetric and skewed empirical distributions, it furnishes an automatic and robust threshold that can be applied across all data sets analysed in the present study.